

Project 22: Ravenhill Coffee Shop Management System

About the Client

Ravenhill Coffee is a specialty coffee roaster and retailer, dedicated to sourcing, roasting and sharing exceptional coffee in a sustainable, respectful and responsible way. We want to establish our first coffee shop in Melbourne CBD starting January 2023 and we need our own POS system to manage our shop.

Ravenhill Coffee is a specialty coffee roaster and retailer, born and based in Melbourne. We are dedicated to sourcing, roasting and making beautiful coffee, and we're committed to doing so in a responsible, sustainable and respectful way.

Ravenhill was founded in 2009 by Fleur Studd and Jason Scheltus. It's hard to fathom now, but at that time, finding fresh, in-season, traceable, high quality coffee in Melbourne (or Australia) was practically impossible. We wanted to help address that and, in doing so, ignite positive change in the industry by redefining what coffee was and could be, and to build a market for, and appreciation of, high quality specialty coffee.

We built our first shop and roastery in the wonderful, bustling Prahran Market. We fell in love with this location for many reasons, the most important being that it allows us to engage with a community of shoppers who are seeking out quality produce, and who care about seasonality and provenance.

We realised early on that we wanted to deliver Melbourne's best coffee experience and to do this, we needed to put all our energy into sourcing, roasting and sharing the very best coffee we could find. Our goal was to make these coffees accessible and exciting, easy to understand and appreciate, and simple to brew and enjoy.

Project Brief & Business Problem Specifications:

This project should provide an automated platform for coffee shop, so that Ravenhill Coffee could handle their daily sales transactions in a more efficient and productive manner. The application should provide the user with the capability to save, manage, and retrieve the sales transactions associated with the specified business. The management of the Ravenhill Coffee Shop is the only ones who have access to this POS for their shop. It is necessary for the

management to log in to the system using their current valid system credentials in order for them to be granted access to the features and functions that are available. The Administrator, the Staff, and the Cashier are the three distinct sorts of user roles that are able to log in and use this system. The Admin user should have the ability to access and administer all of the features and operations of the system, including the management of users and system information. This capability should only be available to the Admin user. The Cashier users should only access the sales transactions that they themselves have generated, whereas the Staff users should have restricted capabilities. Additionally, the project should provide a daily sales report as well as a printable receipt.

The user of the system must provide his or her user credential in order to access and use the features and functionalities of this straightforward project because it will possess a secure login feature. This means that the user can only access and use the features and functionalities of this straightforward project. In addition to storing the sales transactions, the system should also have the capability of storing a list of the many types of water jars and containers, as well as their prices.

System Modules

- User and Access Management
- Customer Management
- Menu Management
- Product Customisation Management
- Table Management
- Reservation Management
- Order Management
- Order Preparation and Tracking
- Payment and Receipt Management
- Discount and Promotion Management
- Loyalty Program Management
- Inventory Management
- Recipe Management

- Supplier and Purchase Management
- Employee and Attendance Management
- Customer Feedback Management
- Dashboard, Reporting and Administration

User Modules (User Frontend):

Developers need to research and discuss with the client to finalise the modules and requirements.

Input Data and Validation Requirements

- Avoiding errors in data
- Controlling amount of input
- Integration of all the modules/forms in the system.
- Preparation of the test cases.
- Preparation of the possible test data with all the validation checks.
- Black-box/White-box testing.
- Recording of all the reproduced errors.
- Modifications should be done for the errors found during testing.
- Prepare the test result scripts after rectification of the errors
- Functionality of the entire modules/forms.
- Validations for user input.

- Checking of the Coding standards to be maintained during coding.
- Testing the modules with all the possible test data.
- Testing of the functionality involving all type of calculations.
- Commenting standard in the source files.
- Easy & fast retrieval of information.
- Well-designed reports.
- Decrease the load of the person involve in existing manual system.
- Access of any information individually.
- Work become very speedy.
- Easy to update information

UI Design

User Interface Design is concerned with the dialogue between a user and the computer. It is concerned with everything from starting the system or logging into the system to the eventually presentation of desired inputs and outputs. The overall flow of screens and messages is called a dialogue.

UI Design Requirements

1. The system user should always be aware of what to do next.
2. The screen should be formatted so that various types of information, instructions and messages always appear in the same general display area.

3. Message, instructions or information should be displayed long enough to allow the system user to read them.
4. Use display attributes sparingly.
5. Default values for fields and answers to be entered by the user should be specified.
6. A user should not be allowed to proceed without correcting an error.
7. The system user should never get an operating system message or fatal error.

The aim of proposed system is to develop a system of improved facilities. The proposed system can overcome all the limitations of the existing system. The system provides proper security and reduces the manual work.

- Security of data.
- Ensure data accuracy's
- Proper control of the higher officials.
- Minimize manual data entry.
- Minimum time needed for the various processing.
- Greater efficiency.
- Better service.
- User friendliness and interactive.
- Minimum time required.

Functional Requirements

1. User and Access Management

FR1. User account creation

FR2. User login

FR3. Role-based access control

FR4. Password reset and profile update

2. Customer Management

FR5. Customer registration

FR6. Customer information storage

FR7. Customer search and update

FR8. Customer history viewing

3. Menu Management

FR9. Menu category creation

FR10. Menu item creation

FR11. Menu item update and deletion

FR12. Menu item availability management

4. Product Customisation Management

FR13. Customisation option creation

FR14. Product customisation selection

FR15. Additional customisation pricing

FR16. Customisation summary display

5. Table Management

FR17. Table registration

FR18. Table information management

FR19. Table status management

FR20. Table assignment

6. Reservation Management

FR21. Reservation creation

FR22. Reservation detail recording

FR23. Table availability checking

FR24. Reservation update and cancellation

7. Order Management

FR25. Order creation

FR26. Order type selection

FR27. Order number generation

FR28. Order total calculation

8. Order Preparation and Tracking

FR29. Barista order display

FR30. Kitchen order display

FR31. Order status update

FR32. Order status tracking

9. Payment and Receipt Management

FR33. Payment processing

FR34. Payment method selection

FR35. Payment status management

FR36. Receipt generation

10. Discount and Promotion Management

FR37. Discount creation

FR38. Promotion type management

FR39. Promotion period management

FR40. Promotional code validation

11. Loyalty Program Management

FR41. Loyalty account registration

FR42. Loyalty point allocation

FR43. Loyalty point redemption

FR44. Loyalty transaction history

12. Inventory Management

FR45. Inventory item registration

FR46. Inventory detail management

FR47. Stock transaction recording

FR48. Stock alert generation

13. Recipe Management

FR49. Recipe creation

FR50. Ingredient assignment

FR51. Automatic stock deduction

FR52. Out-of-stock menu restriction

14. Supplier and Purchase Management

FR53. Supplier registration

FR54. Purchase order creation

FR55. Purchase order tracking

FR56. Purchased stock receiving

15. Employee and Attendance Management

FR57. Employee registration

FR58. Employee schedule management

FR59. Employee clock-in and clock-out

FR60. Attendance approval

16. Customer Feedback Management

FR61. Customer feedback submission

FR62. Order-based feedback linking

FR63. Feedback response management

FR64. Customer rating calculation

17. Dashboard, Reporting and Administration

FR65. Role-based dashboard display

FR66. Report generation

FR67. Report filtering

FR68. Report export and audit logging

Non-functional Requirements

1. Performance Requirements

NFR01. Maximum three-second page loading time

NFR02. Maximum two-second database response time

NFR03. Real-time order status updates

NFR04. Fast payment transaction processing

NFR05. Maximum ten-second report generation time

NFR06. Support for at least 100 concurrent users

2. Security Requirements

NFR07. Secure user authentication

NFR08. Password hashing

NFR09. Role-based access control

NFR10. SQL injection protection

NFR11. Cross-site scripting protection

NFR12. Cross-site request forgery protection

NFR13. Secure session management

NFR14. Automatic session timeout

NFR15. Login attempt restriction

NFR16. HTTPS communication

NFR17. Secure file upload validation

NFR18. Sensitive data access restriction

3. Privacy Requirements

NFR19. Customer data confidentiality

NFR20. Employee data confidentiality

NFR21. Customer consent management

NFR22. Promotional communication opt-out

NFR23. Minimum personal data collection

NFR24. Authorised personal data modification

4. Usability Requirements

NFR25. Simple user interface

NFR26. Consistent navigation

NFR27. Clear form labels

- NFR28.** Understandable error messages
- NFR29.** Minimal order-entry steps
- NFR30.** Search and filtering support
- NFR31.** Confirmation before critical actions
- NFR32.** User-friendly dashboard

5. Audit and Monitoring Requirements

- NFR33.** User activity logging
- NFR34.** Login attempt logging
- NFR35.** Order modification logging
- NFR36.** Refund activity logging
- NFR37.** Inventory adjustment logging
- NFR38.** Audit log search and filtering
- NFR39.** Audit log access restriction

Apart from other NFRs following NFRs must be implemented in the system:

Less clicks and keyboard interactions

Consistent look between website/app and the style in store

Account/phone number verification

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.